

CPE 470/670 Fall 2025
Final Project Demo and Report

Final Demo Details: 10:15 am - 12:15 pm December 17, 2025 in room SEM 342
Project Report Due: 11:59 pm December 17, 2025

Instructor: Parikshit Maini

Maximum Demo Score : 200 points; Report: 50 points

The final project demo must be completed and submitted as a team. Each student will submit a report that includes a write up of all of the components used for this final demo and will include the code used. Each team will write a single report. However, each student will write a one-pager on their individual learning experience from the project, and give a peer rating for each team member out of 10 and include it in their submission of the project report. The report must be submitted as a single PDF file.

Each team member must be fully conversant with all parts of the deliverable. In addition to the submission, each team member will be individually evaluated in an oral exam at the time of the demo that will contribute to 50% of the overall individual grade.

You are not permitted to seek help from people outside the course or AI based tools. You are permitted to refer to class notes, lecture materials, books and other online teaching resources. You are permitted to use standard Python library functions. In case of doubts or questions, please reach out to the Instructor or TAs.

Question: You are given an environment as shown in Figure 1. The size of the environment, walls obstructing robot movement and the placement of the opening from room ABCD is already known. The virtual wall is shown only to mark the area of the demo, they are not physical walls. The environment is equipped with ArUco tags to help localize and navigate. All ArUco tags will be placed at a height of 8.5 cm. All ArUco tags are selected from DICT_4x4_50 dictionary pattern. The origin of the frame of reference for this arena is located at A with positive X and Y axis directions shown in the figure 1.

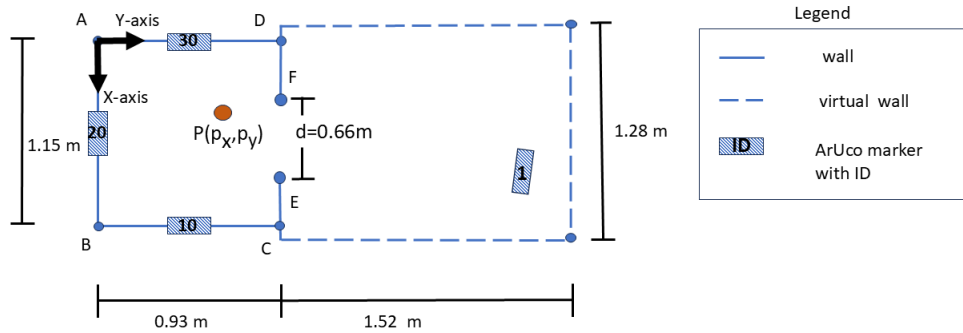


Figure 1: Environment setup for the final demo

Three walls of the enclosed space ABCD will have an ArUco tag with known IDs as shown in the figure. The ArUco tags will be placed at the center of the walls for side AB (marker ID 20), side BC (marker ID 10) and side AD (tag ID 30). Side CD has an opening in the wall shown by EF. $\mathbf{P}$ is a point inside ABCD with coordinates (P_x, P_y) in the arena frame of reference. An ArUco tag with ID 1 will be placed at an unknown location within the area marked by dashed lines outside the room ABCD facing the opening EF.

For this demo, the robot will start at a random location in the room ABCD. At the beginning of your demo, your system must take an input from the user for the location coordinates of point $\mathbf{P}$. This input will be

two positive real numbers with two decimal places marking the location of point P from origin A in meters. Sample coordinates for point **P** for testing can be used at (0.59, 0.46).

The goals for the robot are:

1. navigate to point P and stop for 3 seconds. The robot's center must reach the desired point and
2. navigate out of room ABCD via the opening EF
3. drive into the ArUco tag with the ID 1
4. not to hit any walls of the arena during the demo

You are welcome to use any strategy for your robot to self-navigate to achieve the intended goals. You can use any sensor used in class including LiDAR, camera, wheel encoders and/or robot odometry estimates. You are required to use atleast one sensor. You will receive 50 points for each of the four goals achieved successfully.

Bonus 1: The demo will be timed. The team that achieves the goal in the least time will receive 20 bonus points.

Bonus 2: If you use more than one sensor to achieve the goals for the task you will receive 20 bonus points for every additional sensor used.